EAST WEST UNIVERSITY

Department of Computer Science and Engineering
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Lab Assignment-01
(PL/SQL)

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Tasks

Lab Task # 01 (Create table):

Code:

```
CREATE TABLE instructor_2022360011 (
    id NUMBER PRIMARY KEY,
    name VARCHAR2(50),
    dept_name VARCHAR2(50),
    salary NUMBER
);
```

Output:

```
Table created.
```

Lab Task # 02 (Inserting data into a table):

Code:

```
INSERT INTO instructor_2022360011
VALUES (10101, 'Srinivasan', 'Comp. Sci.', 65000);

INSERT INTO instructor_2022360011
VALUES (12121, 'Wasif', 'Finance', 90000);

INSERT INTO instructor_2022360011
VALUES (15151, 'Mozart', 'Music', 40000);

INSERT INTO instructor_2022360011
VALUES (22222, 'Einstein', 'Physics', 95000);

INSERT INTO instructor_2022360011
VALUES (32343, 'El Said', 'History', 60000);

INSERT INTO instructor_2022360011
VALUES (33456, 'Goblin', 'Physics', 87000);

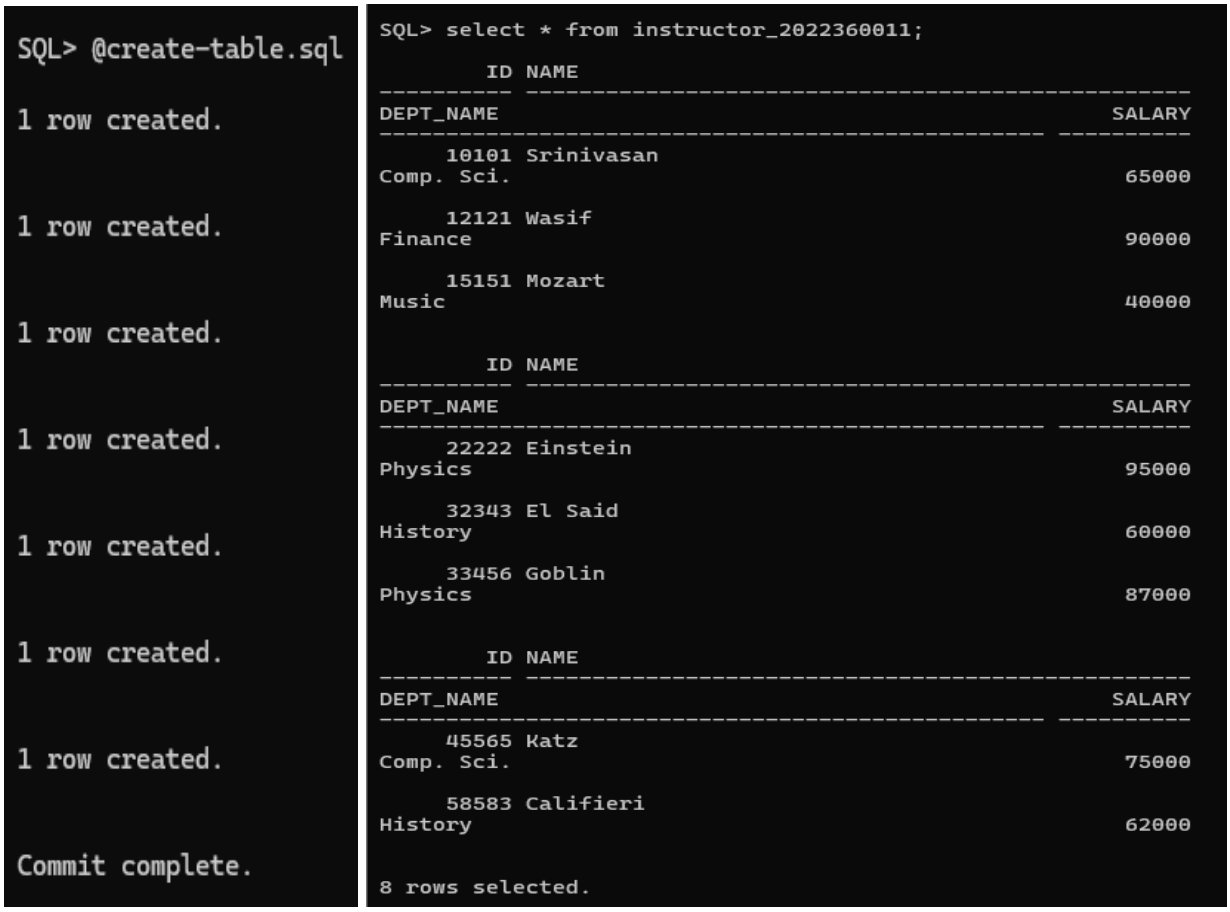
INSERT INTO instructor_2022360011
```

```
VALUES (45565, 'Katz', 'Comp. Sci.', 75000);

INSERT INTO instructor_2022360011
VALUES (58583, 'Califieri', 'History', 62000);

COMMIT;
```

Output:



```
SQL> @create-table.sql

1 row created.


1 row created.


1 row created.


1 row created.


1 row created.


1 row created.


1 row created.


Commit complete.
```

```
SQL> select * from instructor_2022360011;

        ID NAME
---------- --------------------------------------------------
DEPT_NAME                                           SALARY
-------------------------------------------------- ----------
     10101 Srinivasan
Comp. Sci.                                           65000

     12121 Wasif
Finance                                              90000

     15151 Mozart
Music                                                40000


        ID NAME
---------- --------------------------------------------------
DEPT_NAME                                           SALARY
-------------------------------------------------- ----------
     22222 Einstein
Physics                                              95000

     32343 El Said
History                                              60000

     33456 Goblin
Physics                                              87000


        ID NAME
---------- --------------------------------------------------
DEPT_NAME                                           SALARY
-------------------------------------------------- ----------
     45565 Katz
Comp. Sci.                                           75000

     58583 Califieri
History                                              62000

8 rows selected.
```

Lab Task # 03: (PL/SQL Cursor and %TYPE Practice)

3.1 Write a PL/SQL block using implicit cursor to update the salary of instructor with id =15151 by increasing it by 500. Use SQL%ROWCOUNT to print how many rows were updated.

Code:

```
BEGIN
UPDATE instructor_2022360011
SET salary = salary + 500
```

```
WHERE id = 15151;
DBMS_OUTPUT.PUT_LINE('Rows updated: ' || SQL%ROWCOUNT);
END;
```

Output:

```
SQL> @1.sql
  7  /
Rows updated: 1

PL/SQL procedure successfully completed.

SQL> select * from instructor_2022360011;

        ID NAME
---------- --------------------------------------------------
DEPT_NAME                                            SALARY
-------------------------------------------------- ----------
     10101 Srinivasan
Comp. Sci.                                            65000

     12121 Wasif
Finance                                               90000

     15151 Mozart
Music                                                 40500
```

3.2 Declare variables for name and salary using %TYPE. Write a PL/SQL block to select the details of the instructor with id = 22222 into these variables and print them.

Code:

```
DECLARE
v_name instructor_2022360011.name%TYPE;
v_salary instructor_2022360011.salary%TYPE;
BEGIN
SELECT name, salary
INTO v_name, v_salary
FROM instructor_2022360011
WHERE id = 22222;
DBMS_OUTPUT.PUT_LINE('Name: ' || v_name ||', Salary: ' || v_salary);
END;
```

Output:

```
SQL> @2.sql
 11  /
Name: Einstein, Salary: 95000

PL/SQL procedure successfully completed.

SQL>
```

3.3 Write a PL/SQL block using exception handling to select instructor details for id = 99999. If no data is found, print “Instructor not found”.

Code:

```
DECLARE
v_name instructor_2022360011.name%TYPE;
v_salary instructor_2022360011.salary%TYPE;
BEGIN
SELECT name, salary
INTO v_name, v_salary
FROM instructor_2022360011
WHERE id = 99999;
DBMS_OUTPUT.PUT_LINE('Name: ' || v_name ||', Salary: ' || v_salary);
EXCEPTION
WHEN NO_DATA_FOUND THEN
DBMS_OUTPUT.PUT_LINE('Instructor not found');
END;
```

Output:

```
SQL> @3.sql
 15  /
Instructor not found

PL/SQL procedure successfully completed.

SQL>
```

3.4 Write a PL/SQL block using an explicit cursor to fetch and print the details of all instructors from the table. Include steps for declaring, opening, fetching in a loop, and closing the cursor.

Code:

```
DECLARE
v_id instructor_2022360011.id%TYPE;
v_name instructor_2022360011.name%TYPE;
v_dept instructor_2022360011.dept_name%TYPE;
v_salary instructor_2022360011.salary%TYPE;
CURSOR c_instructor IS
SELECT id, name, dept_name, salary
FROM instructor_2022360011;
BEGIN
OPEN c_instructor;
LOOP
FETCH c_instructor
INTO v_id, v_name, v_dept, v_salary;
EXIT WHEN c_instructor%NOTFOUND;
DBMS_OUTPUT.PUT_LINE('ID: ' || v_id ||', Name: ' || v_name ||', Department: ' || v_dept
||', Salary: ' || v_salary);
END LOOP;
DBMS_OUTPUT.PUT_LINE('Total rows fetched: ' || c_instructor%ROWCOUNT);
CLOSE c_instructor;
END;
```

Output:

```
SQL> @4.sql
  20  /
ID: 10101, Name: Srinivasan, Department: Comp. Sci., Salary: 65000
ID: 12121, Name: Wasif, Department: Finance, Salary: 90000
ID: 15151, Name: Mozart, Department: Music, Salary: 40500
ID: 22222, Name: Einstein, Department: Physics, Salary: 95000
ID: 32343, Name: El Said, Department: History, Salary: 60000
ID: 33456, Name: Goblin, Department: Physics, Salary: 87000
ID: 45565, Name: Katz, Department: Comp. Sci., Salary: 75000
ID: 58583, Name: Califieri, Department: History, Salary: 62000
Total rows fetched: 8

PL/SQL procedure successfully completed.

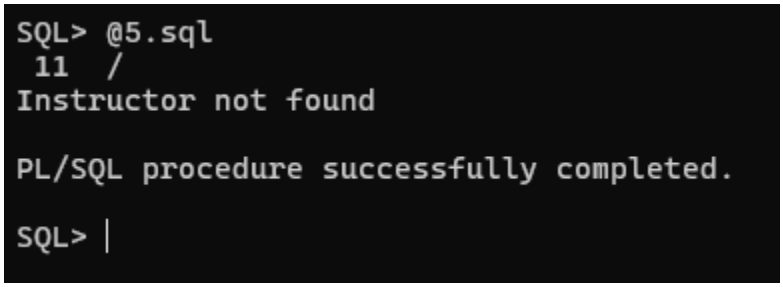
SQL>
```

3.5 Write a PL/SQL block to delete an instructor with id = 12345. Use SQL%NOTFOUND to print "Instructor not found" if no record was deleted, otherwise print "Instructor deleted successfully".

Code:

```
BEGIN
DELETE FROM instructor_2022360011
WHERE id = 12345;
IF SQL%NOTFOUND THEN
DBMS_OUTPUT.PUT_LINE('Instructor not found');
ELSE
DBMS_OUTPUT.PUT_LINE('Instructor deleted successfully');
END IF;
END;
```

Output:



3.6 Write a PL/SQL block using an explicit cursor to fetch all instructors whose salary is less than 60000, update their salary by adding 1000, and count how many instructors were updated. Print the total count after processing.

Code:

```
DECLARE
v_id instructor_2022360011.id%TYPE;
v_salary instructor_2022360011.salary%TYPE;
v_count NUMBER := 0;
CURSOR c_instructor IS
SELECT id, salary
FROM instructor_2022360011
WHERE salary < 60000;
BEGIN
OPEN c_instructor;
LOOP
```

```
FETCH c_instructor INTO v_id, v_salary;
EXIT WHEN c_instructor%NOTFOUND;
UPDATE instructor_2022360011
SET salary = salary + 1000
WHERE id = v_id;
IF SQL%FOUND THEN
v_count := v_count + 1;
DBMS_OUTPUT.PUT_LINE('Salary updated for ID: ' || v_id);
END IF;
END LOOP;
CLOSE c_instructor;
DBMS_OUTPUT.PUT_LINE('Total instructors updated: ' || v_count);
END;
```

Output:

